



Context-Aware IoT-Based Field Monitoring for Sri Lankan Paddy Cultivation Decision Support System

Project ID :R26-IT-066



The Real-World Problem

1.8 Million Households at Risk

Sri Lankan paddy farming households face chronic income instability and reduced productivity due to unpredictable weather, soil imbalances, and fungal diseases.

Inefficient Decision-Making

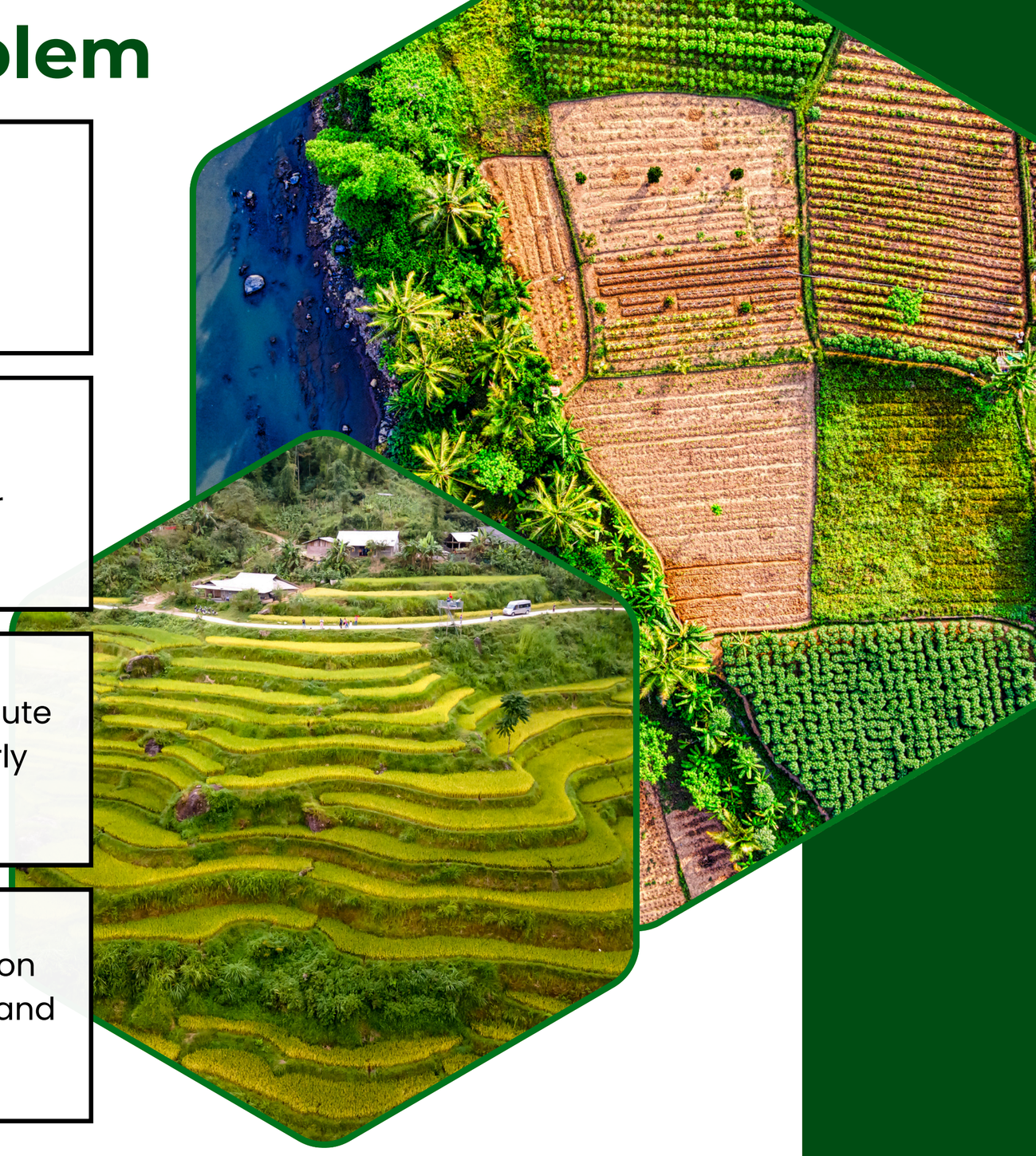
Farmers rely on traditional knowledge and fixed guidelines rather than real-time field data—leading to improper fertilizer timing and late disease detection.

>30% Yield Loss in Severe Seasons

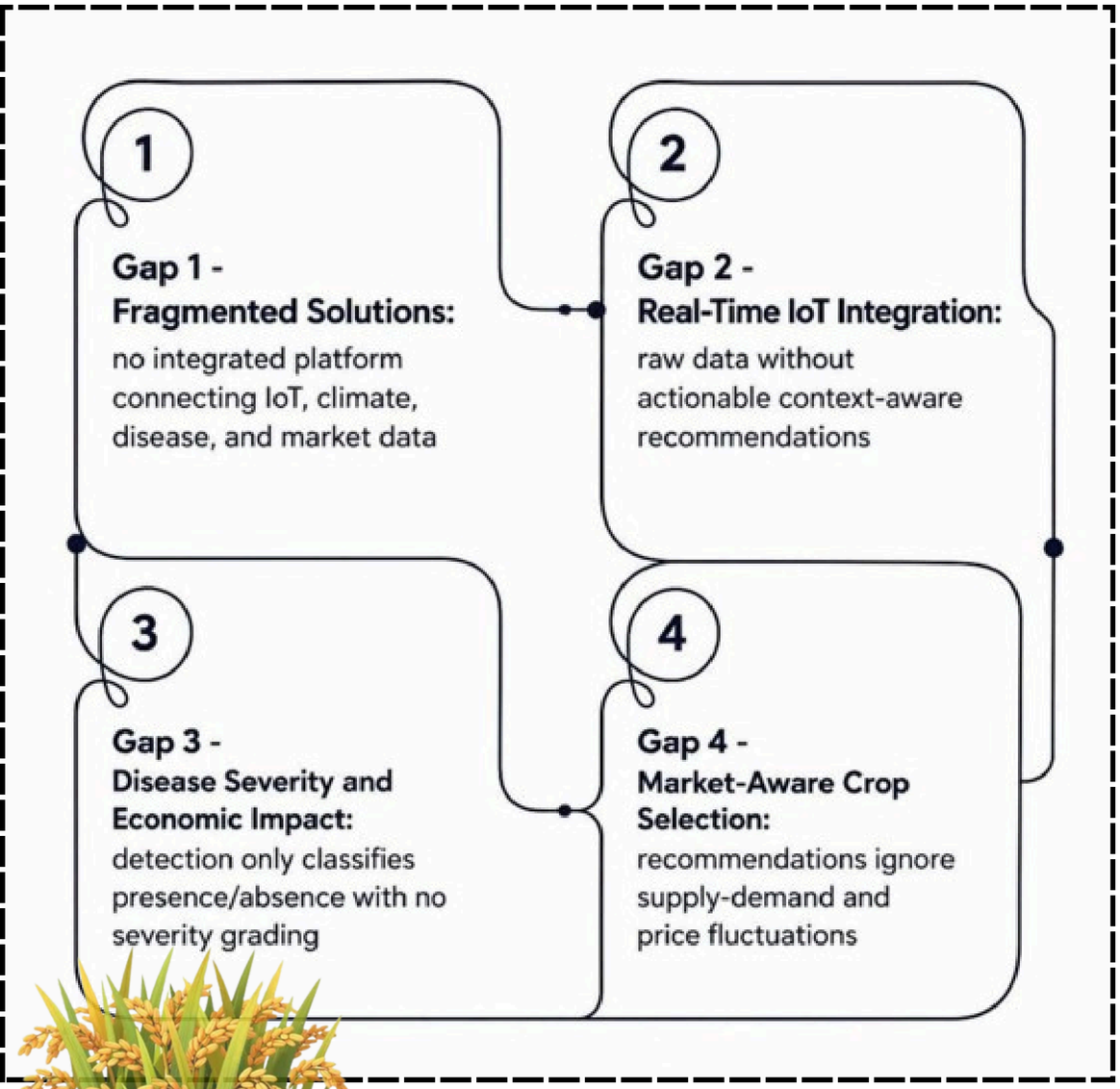
Rice Blast, Brown Spot, and uninformed crop selection contribute to yield losses exceeding 30% when conditions are not properly monitored.

No Integrated Solution

Existing systems are fragmented—IoT monitoring lacks decision support, disease detection provides no severity assessment, and crop recommendations ignore market economics.



Research Gaps Identified



Why These Gaps Matter

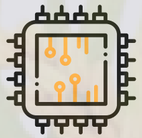
A systematic review of 9+ scholarly sources and existing Sri Lankan platforms revealed that no single platform addresses all four dimensions of paddy farm decision-making simultaneously.

- **Gap 1 — Fragmented Solutions:** No platform integrates IoT, climate, disease, and market intelligence
- **Gap 2 — Real-Time IoT Integration:** Raw sensor data without actionable, localized recommendations
- **Gap 3 — Disease Severity & Economic Impact:** No severity grading or yield-loss forecasting
- **Gap 4 — Market-Aware Crop Selection:** Recommendations ignore price fluctuations and supply-demand

Research Objectives

Overall Goal: Develop an integrated, context-aware IoT-based decision support system enabling Sri Lankan paddy farmers to make data-driven, economically optimized cultivation decisions.

Component 1 — IoT Field Monitoring



Continuously monitor soil moisture, pH, NPK, temperature and generate real-time fertilization alerts

Component 2 — Crop Recommendation



Integrate IoT data, climate forecasts, and market price trends to recommend economically viable crops

Component 3 — Disease Detection & Severity



Detect Rice Blast and Brown Spot with multi-class severity grading (0–4 scale) and management recommendations

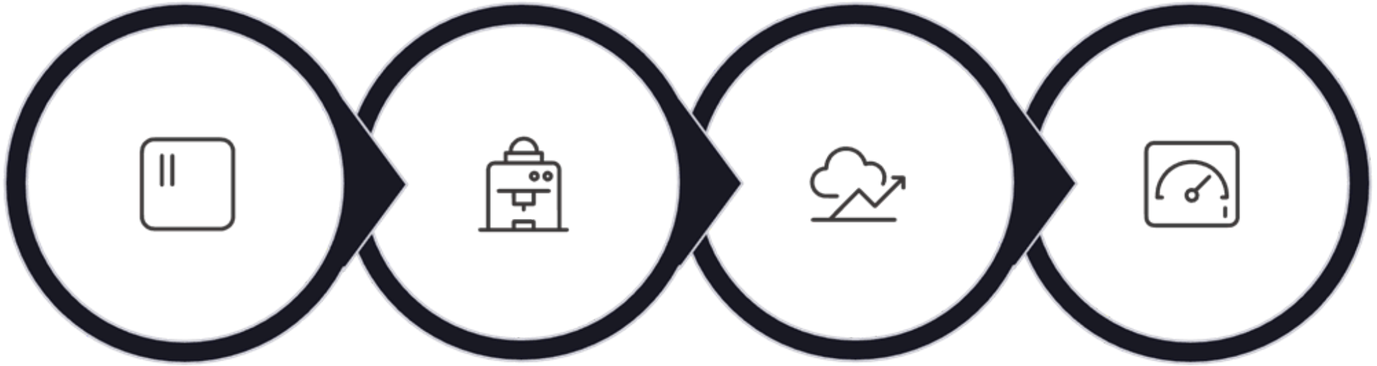
Component 4 — Harvest Yield Estimation



Predict yield under three scenarios (healthy, untreated disease, treated disease) to enable evidence-based treatment decisions

Proposed Integrated Solution

Vision: A unified, farmer-centric decision support platform that transforms fragmented data into actionable intelligence.



Field Monitoring

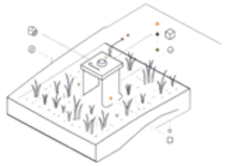
Intelligent Analysis

Scenario Recommendations

Farmer Dashboard



Four Integrated Research Components



Component 1 — IoT Field Monitoring

Real-time soil sensing (moisture, pH, NPK, temperature, humidity) → Fertilization timing alerts *(IT22178718)*



Component 2 — Crop Recommendation

IoT data + climate APIs + market price forecasts → Economically viable crop suggestions *(IT21811104)*



Component 3 — Disease Detection & Severity

Paddy leaf images + IoT environmental data → Disease type + severity grade (0–4) + management recommendations *(IT22261632)*

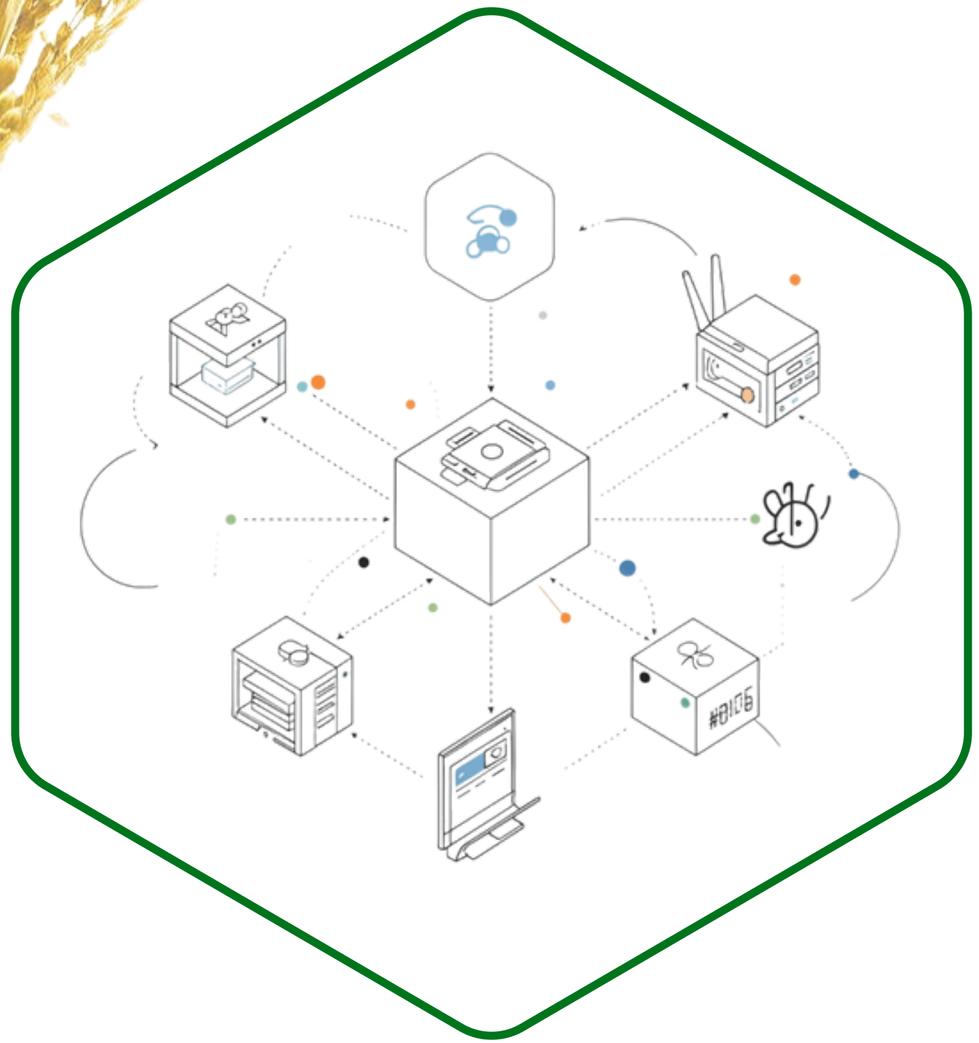


Component 4 — Harvest Yield Estimation

IoT data + disease severity + IRRI yield-loss models → Three-scenario yield forecasts *(IT22266446)*

Key Integration: All components share a central cloud database and REST APIs, enabling cross-component data flow and a unified farmer dashboard

System Architecture Overview



Data Sources

IoT field sensors → Weather APIs → Paddy leaf images → Market price databases → IRRI yield-loss reference data

Processing Layer

AWS IoT Core & Analytics → Machine learning models (Random Forest, XGBoost, EfficientNet-B4 CNN)

Integration Layer

FastAPI REST endpoints enable cross-module communication and scenario-based data flow

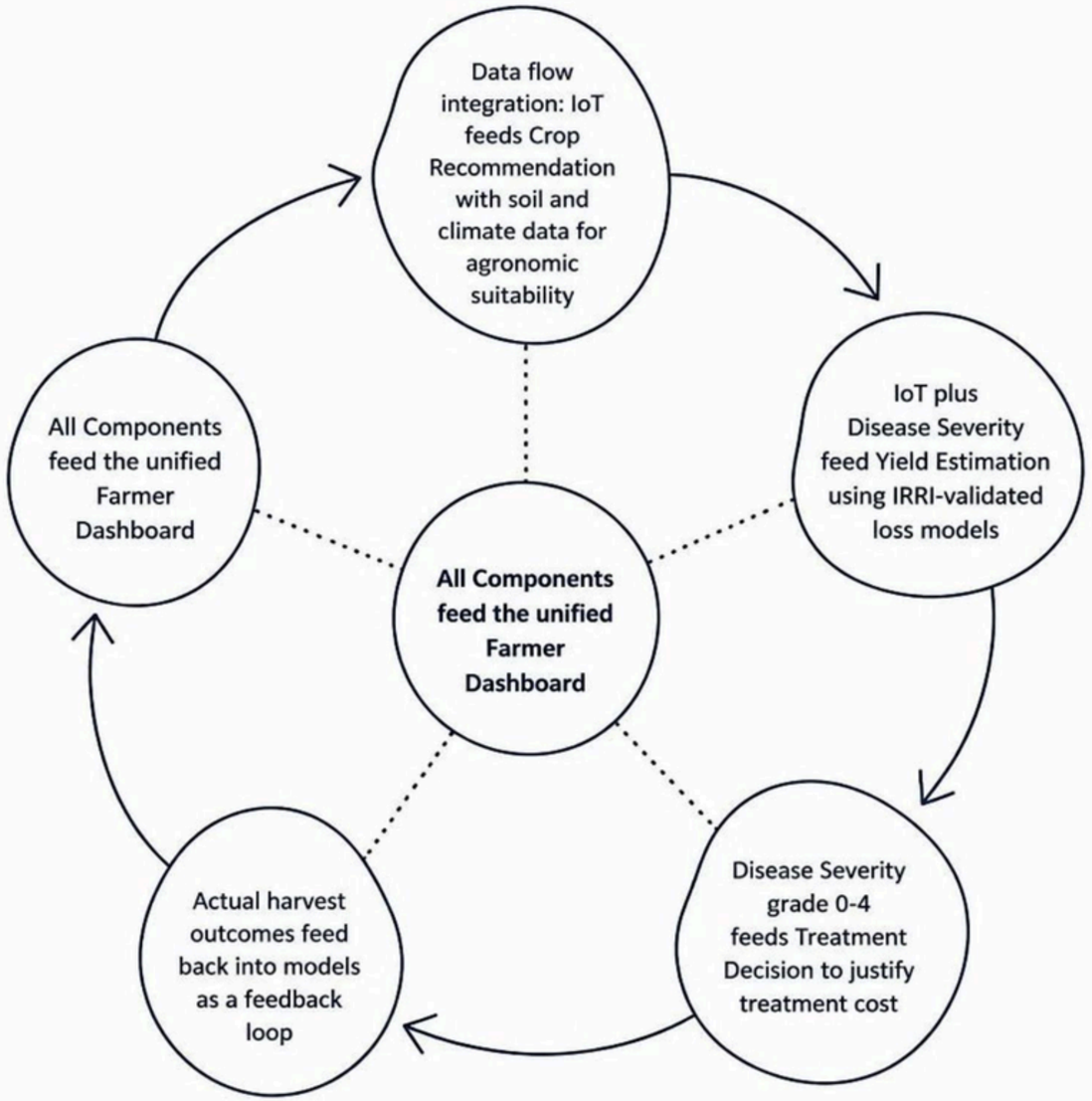
Output Layer

React.js web dashboard — fertilization alerts, crop recommendations, disease severity, yield scenarios, and financial impact

Storage

PostgreSQL + AWS S3 (encrypted images) + historical yield and market data

Component Integration & Data Flow



How the Components Connect

- **IoT →** Crop Recommendation: Real-time soil and climate data inform agronomic suitability
- **IoT + Disease Severity →** Yield Estimation: Environmental conditions and severity scores feed IRRI-validated yield-loss models
- **Disease Severity → T**reatment Decision: Severity grade (0–4) and yield impact help farmers decide if treatment cost is justified
- **All Components →** Dashboard: Integrated outputs appear on a single farmer interface
- **Feedback Loop:** Actual harvest outcomes feed back into models to improve future predictions

Research Methodology

1

Phase 1

Problem Identification & Literature Review — systematic review of 9+ scholarly sources and existing Sri Lankan platforms

2

Phase 2

Research Gap & Objectives Definition — four specific gaps defined with measurable outcome alignment

3

Phase 3

System Design & Architecture — integrated architecture with four interconnected components and shared data layer

4

Phase 4

Component Development — implemented with localized data (BG 300, AT 362; IRRI SES; DoA guidelines)

5

Phase 5

Integration & Testing — REST API integration; cross-module data flow and system stability testing

6

Phase 6

Evaluation & Validation — objective-aligned metrics; integrated system functionality demonstrated



Data & Technologies

Data Sources

- IoT sensor data (soil moisture, pH, NPK, temperature, humidity)
- Weather APIs (climate forecasts, rainfall, temperature trends)
- Paddy leaf images — locally collected, annotated by DoA-SL officers
- Market price databases (historical and forecasted)
- IRRI Standard Evaluation System (disease-yield loss reference tables)
- Department of Agriculture Sri Lanka guidelines and historical yield data

Technology Stack

Backend

Python · FastAPI · AWS IoT Core · AWS IoT Analytics

Machine Learning

Random Forest · XGBoost · EfficientNet-B4 · CNN

Frontend

Flutter web dashboard

Cloud & Storage

AWS · PostgreSQL · AWS S3 (encrypted)

Evaluation Strategy

Component-Specific Metrics

- **IoT Monitoring:** Alert accuracy, sensor reliability, real-time responsiveness
- **Crop Recommendation:** Recommendation accuracy, economic viability validation against market outcomes
- **Disease Detection:** Classification accuracy ≥91%, severity grading accuracy, inference time <5 seconds, expert reviewer approval ≥80%
- **Yield Estimation:** Base yield prediction accuracy ≥85%, disease-yield loss model R^2 ≥0.80, scenario forecast accuracy



Evaluation Strategy

System-Level Metrics

•API response time and integration stability
•Dashboard usability and farmer accessibility
•End-to-end system performance under real field conditions

Baseline Comparison

Results compared against traditional farmer practices and existing static recommendation systems to demonstrate measurable improvement in yield outcomes and decision quality.



Thank You

